BRAIN-WIDE NEURAL DYNAMICS OF DECISION-MAKING

A large-scale analysis of distributed neural activity across task stages

Authors

Graziella K. Samey, Marie Ange N. Yedoh

Affiliation

Aix-Marseille University, Faculty of Science

Introduction

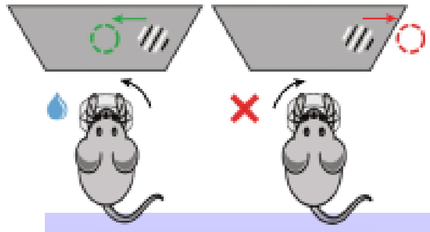
Understanding how the brain transforms sensory input into decisions remains a major challenge in neuroscience.
Previous studies focused on isolated brain regions, leading to fragmented insights.
Recent advances now allow brain-wide recordings, enabling a more complete view of neural activity.
This study investigates whether decision-making is localized or distributed across the brain.

Objective

- Examine how neural activity encodes key decision variables (stimulus, choice, action, reward)
- Determine whether decision-making is localized or distributed
- Identify how different brain regions interact during behavior
- Provide a large-scale dataset for studying brain-wide dynamics

Methodology

1. Mice were trained on a visual decision-making task
   - A stimulus appeared on the left or right
   - Mice responded by turning a wheel
   - Correct choices were rewarded
2. Task difficulty was manipulated using varying stimulus contrast
   - Including 0% contrast trials (no visual information)
   - Probability blocks (80/20) introduced prior expectations
3. Neural activity was recorded using Neuropixels probes
   - Simultaneous recordings across multiple brain regions
   - 139 mice, 699 probe insertions, ~621,000 neurons
4. Behavioral and neural data were analyzed together
   - Four approaches: decoding, single-cell, population dynamics, encoding
5. This allowed identification of how stimulus, choice, movement, and reward are represented across the brain



Results

The results reveal how neural activity evolves across task stages, from sensory processing to decision-making and action.

Sensory proccessing

Neural responses to stimuli are localized within visual cortical areas. These early responses reflect direct sensory input.

Decision formation

Neural activity becomes distributed across multiple brain regions.
Decision-related signals involve interactions between cortical and subcortical areas.

Movement & Reward

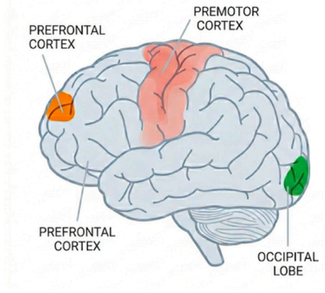
Neural activity becomes widespread across the brain.
Movement and reward engage large-scale neural networks.

Together, these results demonstrate a progressive shift in neural organization, from localized sensory encoding to distributed and brain-wide activity across task stages.

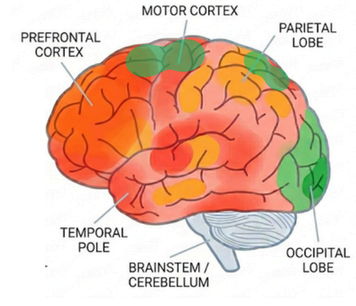
Key Insight
Decision-making is not localized to a single brain region, but emerges from distributed, large-scale neural activity across the brain.

This challenges the traditional view of localized brain function and supports a distributed model of neural processing.

Traditional beliefs



Research's findings



Impact/Discussion

These findings challenge the traditional view of localized brain function by demonstrating that decision-making emerges from distributed neural activity across multiple brain regions.
This network-based organization provides a more comprehensive framework for understanding how complex behaviors are generated in the brain.
Such insights may contribute to a better understanding of neurological disorders, such as Alzheimer's and Parkinson's disease, which involve disruptions in large-scale brain networks.

Conclusion

This study shows that neural activity evolves from localized sensory processing to distributed and brain-wide activation during decision-making and action.
These results support a network-based model of brain function, where cognition arises from interactions across multiple regions rather than isolated modules.

Future directions

Future research could further explore how distributed neural activity supports more complex behaviors and how disruptions in these networks relate to neurological disorders.

INTERNATIONAL BRAIN LABORATORY

Key Sources

International Brain Laboratory. (2025)
A brain-wide map of neural activity during decision-making. Nature.

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